

VASANTH KUMAR

AI Engineer | Agentic AI Developer | AI Security

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PROFILE

Entry-level AI Engineer focused on building AI applications, agentic AI systems, RAG pipelines, and workflow automation, with a growing specialization in AI security. Experienced in developing and deploying agentic AI, LLM-integrated, and automation-driven projects using Python, FastAPI, and modern AI tooling. Seeking entry-level opportunities as an AI Engineer, Agentic AI Developer, or AI Security professional.

TECHNICAL SKILLS

AI Engineering & Agentic AI

Python | FastAPI | GenAI | LLM Integration | RAG | AI Agents | MCP | Tool Calling | Prompt Engineering | Groq API

AI & Workflow Automation

n8n | Python Automation | PowerShell Automation | Workflow Design | Webhooks | REST API Integration

Cloud & Deployment

Vercel | Render | Cloudflare Workers | Docker | AWS Basics

AI Security

Prompt Injection | RAG Security | Agentic AI Security | LLM Security | OWASP AI Security | MITRE ATLAS

Cybersecurity

VAPT | DFIR | Network Security | SOC & SIEM | Threat Analysis | Incident Response

Other — AI Ecosystem

OpenAI | Claude | Groq | AI Model APIs | JSON | Version Control (Git & GitHub) | Linux

PROJECTS

Agent V — Agentic AI Security Gateway

Technologies: Next.js | React | TypeScript | Cloudflare Workers | Cloudflare D1 | Drizzle ORM | Groq API | Tailwind CSS | [Live Demo](#)

- > Developed an agentic AI security gateway that evaluates AI-agent prompts and proposed tool actions using deterministic security rules and Groq-powered analysis.
- > Implemented risk scoring, ALLOW/REVIEW/BLOCK decisions, role-based permission controls, human approval workflows, and traceable audit logging.

RAGExploit — AI & RAG Security Scanner

Technologies: React | Vite | Python | FastAPI | RAG | Groq API | REST API | Docker | Vercel | [Live Demo](#)

- > Developed a full-stack security assessment application with a real RAG pipeline for document parsing, chunking, embeddings, retrieval, and optional LLM-assisted generation.
- > Built five automated scanner modules covering document poisoning, indirect prompt injection, unauthorized retrieval, sensitive-data leakage, and retrieval manipulation with evidence-backed findings and risk scoring.

BlueOrch — SOC Investigation & Response Automation Platform

Technologies: React | TypeScript | Python | FastAPI | PostgreSQL | SQLAlchemy | n8n | Groq | RAG | MCP | REST APIs | [Live Demo](#)

- > Developed an end-to-end automation platform connecting a React interface with FastAPI services for event ingestion, enrichment, AI-assisted investigation, approval, and response workflows.
- > Integrated Groq structured analysis, RAG-based runbook retrieval, n8n orchestration, and 7 audited MCP tools with human-gated response actions and persistent incident data.

V-CyberWar — RED vs BLUE

Technologies: Python | RAG | MCP | Multi-Agent Orchestration | Memory | [Live Demo](#)

- > Building a Red-vs-Blue multi-agent cyber-range centered around a V-Core orchestration layer that coordinates a Red Agent and a Blue Agent within a controlled, isolated environment.
- > Designing RAG- and MCP-based tool access, persistent memory, and a Judge/evaluation component to assess and score agent actions during simulated engagements.

EXPERIENCE

Full Stack Development Intern

Livestream Technology, Coimbatore | June 2025

- > Built responsive web pages using HTML, CSS, JavaScript, and React.js.
- > Connected frontend components with REST APIs to display and manage application data.
- > Used Git and GitHub for version control and collaborated on development tasks.

Independent AI Engineering

Independent Projects | 2026 – Present

- > Building AI applications, RAG systems, agentic workflows, and automation using Python/FastAPI and LLM integrations.
- > Applying AI security concepts while developing and testing AI applications and agent workflows.

EDUCATION

Master of Computer Applications (MCA) | 2026–2028 (Pursuing)

Sai Bharath College | Affiliated to Madurai Kamaraj University

B.Sc. Computer Science (Digital Forensic Science & Cybersecurity) | 2023–2026

Rev. Jacob Memorial Christian College | Affiliated to Madurai Kamaraj University